

Abhirup Majumder

Data Scientist | Gen AI & LLM Developer | Agentic AI Engineer

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SUMMARY

Data Scientist and Generative AI Developer with ~3 years building production-grade LLM, Agentic AI and RAG systems at enterprise scale at Accenture. Proven record turning frontier AI research into measurable business impact across tax automation, financial analytics and intelligent audit. GATE top-5%% national rank and national award winner.

CORE SKILLS

Gen AI & LLM: LangChain, LangGraph, LlamaIndex, AutoGen, ReAct, GPT (OpenAI), Claude, HuggingFace, RAG, Vector Stores, Multi-Agent Systems, LLM Evaluation & Benchmarking, Semantic Search

ML & Statistics: PyTorch, Scikit-learn, XGBoost, LightGBM, Keras, PySpark, EDA, Regression, Forecasting, Statistics

Data & Cloud: Snowflake, Databricks, Delta Lake, AWS (S3, Lambda), Docker, Kubernetes, Airflow, MLflow

Enterprise & SAP: SAP BTP, ABAP, S/4HANA, OData, CDS Views, OAuth 2.0, REST APIs, MS Graph API

Programming: Python, SQL, C++, JavaScript, Bash, Git, CI/CD

EXPERIENCE

Data Scientist & Gen AI Developer

Jul 2023 – Present

Accenture · Computational AI / ML

- Architected Agentic AI workflows (LangGraph, ReAct) to automate U.S. tax determination over large unstructured invoice datasets; deployed RAG pipelines with vector-store integration for context-aware LLM inference, integrating SAP HANA Data Mart and the VERTEX tax engine via REST APIs.
- Engineered GL & Tax classification agents (GPT, Claude) on SAP S/4HANA (ACDOCA) data hosted on SAP BTP with OAuth 2.0; built LLM evaluation and benchmarking frameworks to validate model outputs before production rollout.
- Led an LLM-powered email-automation system (NLP + Microsoft Graph API) using Agentic patterns to autonomously query SAP S/4HANA, **significantly reducing manual workload** for audit and compliance teams.
- Built SAP BTP enterprise ETL pipelines (CDS views, OData) and tuned legacy ABAP code for throughput and data integrity.

ML Research Intern

Oct 2021 – Jul 2022

IIT Kharagpur

- Built end-to-end Python/MySQL ML pipelines on satellite telemetry data; tuned a poacher-detection model contributing to a **50%% reduction in local poaching rates**, with methodology contributed to open-source research.

Machine Learning Intern

Feb 2020 – Jun 2020

Verzeo

- Built supervised ML pipelines for fraud detection, churn prediction and sales forecasting; ETL and REST integrations improved early-engagement prediction accuracy by **200%%**.

SELECTED PROJECTS

- **Jane Street Market Prediction** — High-frequency trading ensemble (MLP + GBDT) over 130+ features with bottleneck-autoencoder denoising and Purged Group Time-Series CV. *PyTorch, LightGBM, XGBoost, Keras*.
- **IoT-Based Smart HVAC System** — Real-time IoT analytics pipeline achieving 25%% energy reduction; 3rd place, ISHRAE All-India National Design Competition. *Python, MQTT, Firebase, Arduino*.

EDUCATION & AWARDS

- **B.Tech, Electrical Engineering** — Institute of Engineering & Management, Kolkata · CGPA 9.08 / 10 · 2019–2023
- **GATE** — Top 5%% nationally (Rank 4014, Score 500) · **ISHRAE** National Design Competition — 3rd Place · **IIT Kharagpur** open-source research contribution